Study Breaks, Perceived Study Efficiency, and Next-Day Retention

STA304H5 | Research Proposal | Fall 2026

Research Objective

This study examines the relationship between study-break habits and perceived learning outcomes among students enrolled in STA304H5 in Fall 2026. We focus on break frequency, average break duration, and gaming during breaks. Perceived efficiency is measured by agreement with the statement that taking a break helps students study more efficiently in the next study period; perceived retention is measured by a self-reported next-day recall rating.

Research Questions

1. Is break frequency associated with students' agreement that breaks improve their efficiency in the next study period?

2. Is average break duration linearly associated with perceived next-day retention?

3. Is gaming during breaks associated with perceived next-day retention after accounting for break duration and study-session length?

Population, Sampling, and Data Collection

The target population is students enrolled in STA304H5 in Fall 2026. An online questionnaire was distributed through the course survey exchange platform using Google Forms. Participation was voluntary, making this a voluntary-response sample. Of the 57 submissions received, two contained survey links in place of course-specific IDs and were excluded, leaving 55 records for further eligibility verification.

The questionnaire collects study-session length, number of breaks, average break duration, weekly study hours, break activities, and student background information. Retention is recorded on a 10-point scale and agreement about efficiency on a seven-point scale. Break activities allow multiple selections.

Statistical Analysis

For RQ1, we will fit a simple linear regression with the efficiency agreement score as the response and break count as the predictor. For RQ2, we will fit a simple linear regression with the retention rating as the response and average break duration, in minutes, as the predictor. Each analysis will test whether the slope differs from zero at the 5% significance level and report the estimated slope with a 95% confidence interval.

For RQ3, we will use multiple linear regression with retention as the response and break duration, study-session length, and a gaming indicator as predictors. The indicator distinguishes respondents who selected gaming from those who reported other break activities. We will also consider break count as an additional predictor. Candidate models will be fitted to the same complete-case sample and compared using adjusted R-squared, nested-model F-tests where applicable, and residual diagnostics.

Missing or invalid values will be excluded only from analyses requiring the affected variables. Cleaning decisions are documented in the codebook and cleaning log. We will report the sample size for each analysis, examine linearity, residual variance and approximate normality, and assess influential observations. A sensitivity analysis will evaluate the effect of the unusually long reported study session.

Scope and Limitations

The rating scales will be treated as approximately numerical for regression analysis. Both outcomes reflect students' perceptions rather than measured academic performance. Voluntary participation may introduce selection bias and limit generalizability to the course population. The observational design supports conclusions about associations, but does not establish causal effects of study breaks.

Questionnaire: Google Form